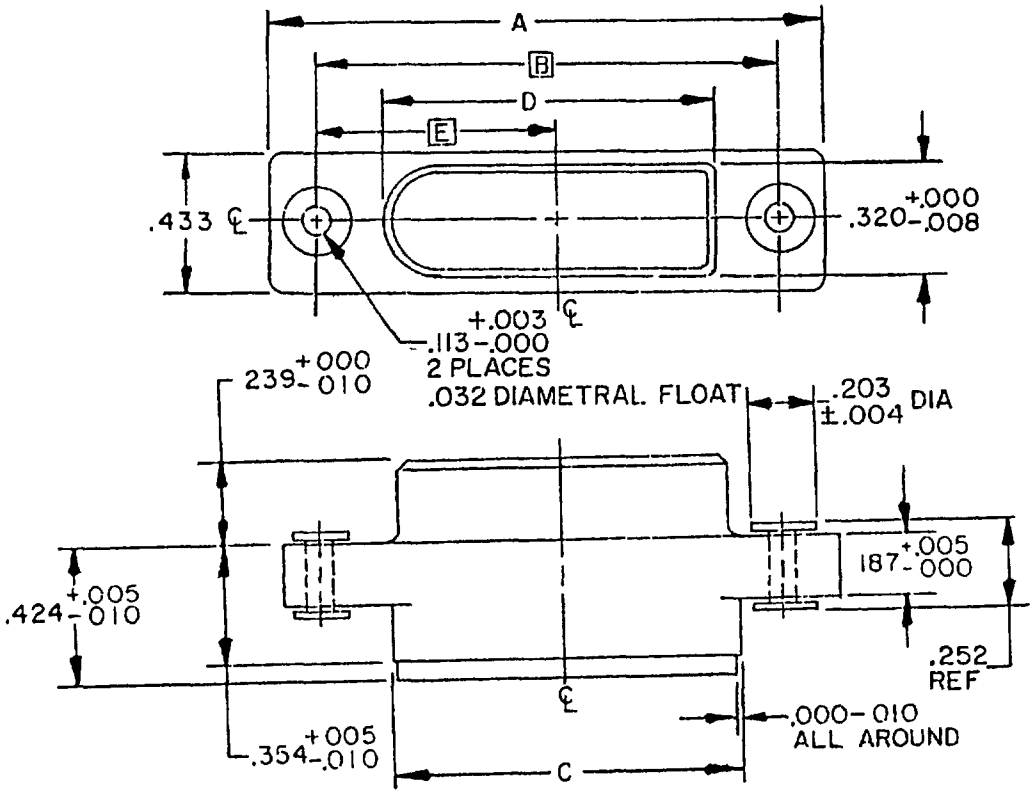
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, SOCKET RECEPTACLE, ELECTRICAL, POLARIZED (.090 SPACING), REMOVABLE CRIMP CONTACTS

© This specification inactive for new design after 1 October 1986

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-55302.



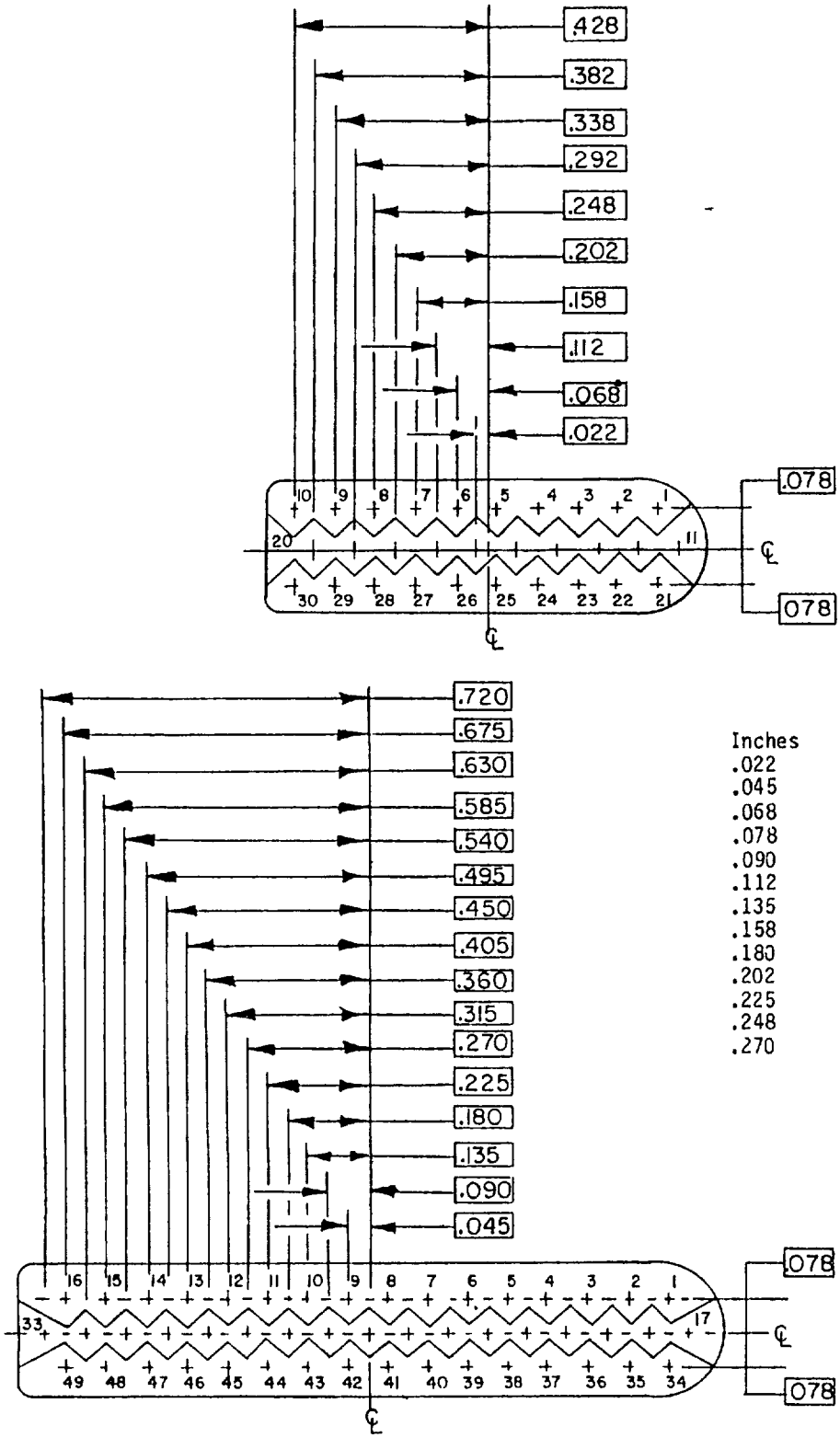
Inches	mm
.003	0.08
.004	0.10
.005	0.13
.008	0.20
.010	0.25
.032	0.81
.113	2.87
.187	4.75
.203	5.16
.239	6.07
.252	6.40
.320	8.13
.354	8.99
.424	10.77
.433	11.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Nominal spacing between any two adjacent socket contacts shall be .090 ±.003 (2.29 ±0.08 mm).
4. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
5. Number 4 round head screw (corrosion resisting steel) recommended for mounting.
6. For dimensions A through E, see table II.
7. A quantity of MS27491-22D socket crimp contacts consisting of the normal complement plus one spare contact for connector arrangements having 26 contacts or less and two spares for arrangements having over 26 contacts shall be supplied with each connector.
8. Quality assurance provisions for MS27491-22D socket are in accordance with MIL-C-38999.

FIGURE 1. Connector, receptacle, .090 (2.29 mm), contact spacing, removable crimp socket.

© denotes changes

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Inches	mm	Inches	mm
.022	0.56	.292	7.42
.045	1.14	.315	8.00
.068	1.73	.338	8.59
.078	1.98	.360	9.14
.090	2.29	.382	9.70
.112	2.84	.405	10.29
.135	3.43	.428	10.87
.158	4.01	.450	11.43
.180	4.57	.495	12.57
.202	5.13	.540	13.72
.225	5.72	.585	14.86
.248	6.30	.630	16.00
.270	6.86	.675	17.15
		.720	18.29

NOTES:
1. Contacts 1, 20, and 21 (Top of Page) are identified on the front face of the connector and contacts 1, 10, 11, 20, 21 and 30 are identified on the rear face per the insert arrangement.
2. Contacts 1, 16, and 49 (Bottom of Page) are identified on front face of the connector and contacts and all contacts are identified on the rear face per the arrangement.

FIGURE 2. Insert arrangement for .090 (2.29 mm) contact spacing connectors (Female engaging face).

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table II.

Material: In accordance with MIL-C-55302, except: Body - reinforced epoxy resin or equivalent.

Plating: In accordance with MS27491-22D.

Contact identification: Contact locations are identified numerically as shown on figure 2.

Mating and unmating: The maximum mating force, in pounds, shall be 0.5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Contact engagement and separation forces: When tested as specified herein, the engaging and separation forces shall be within the applicable limits specified in table I. Cylindrical steel or tungsten carbide test pins having spherical tips and a surface finish not exceeding 3 microinches roughness for steel pins or 10 microinches for tungsten carbide pins, shall be engaged with and separated from the socket contacts. The forces necessary to insert the maximum diameter pin and withdraw the minimum diameter pin shall be measured and shall meet the requirements specified in table I. The depth of insertion shall be not less than 0.205 inch measured from the face of the insert of 0.140 inch measured from the end of the socket body, but the pins shall not bottom in the socket.

TABLE I. Engaging and separating forces.

Mating end size	Engaging test pin diameter in inches	Maximum engagement force (ounces)	Separating test pin diameter in inches	Minimum separation force (ounces)
22	0.0305 +0.0002 -0.0000	12	0.0295 +0.0000 -0.0002	0.75

Contact resistance: With a test current of 3 A dc, the average resistance of all contact pairs measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Oversize pin: The connectors shall exclude a .038 minimum diameter pin.

Dielectric withstanding voltage:

Sea level: 1,000 V rms, 60 Hz, ac.

High altitude: 275 V rms, 60 Hz, ac.

Contact current rating: 3.0 amperes.

Insert arrangement: See figure 2.

Mating connectors: See MIL-C-55302/67 and /69.

Contact insertion an removal tool: See MIL-C-55302/79.

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Quality conformance inspection:

Group A inspection: Delete mating and unmating.

Group B inspection: Add mating and unmating.

Part number: M55302/68-(dash number from table II).

TABLE II. Dash number and dimensions.

Dash number	Dimensions 1/				
	+.010 A (.25) -.000	B	+.010 C (.25)	+.000 D -.008 (.20)	E
01	1.661 (42.19)	1.376 (34.95)	1.047 (26.59)	.986 (25.04)	.688 (17.48)
02	2.246 (57.05)	1.960 (49.78)	1.632 (41.45)	1.571 (39.90)	.980 (24.89)

1/ Metric equivalent are given for general information only. Millimeters are in parentheses.

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 17, 85
DLA - ES

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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